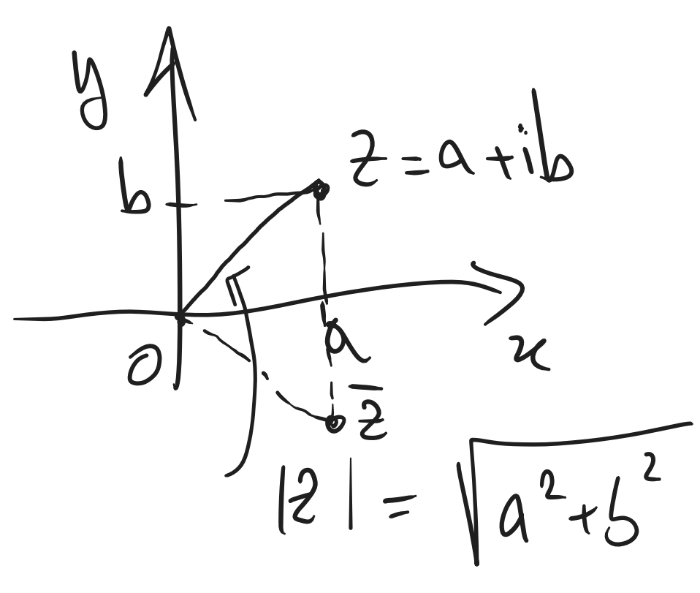
$z = a + ib \in \mathbb{C}$ $\qquad a, b \in \mathbb{R}$

$a = \mathrm{Re}(z) \quad b = \mathrm{Im}(z)$



$|z| = \sqrt{a^2+b^2}$

operazioni in $\mathbb{C}$: $\boxed{i^2 = -1}$

$\bar z = a - ib$
↑ coniugato di z

$\underbrace{z\bar z = |z|^2}$ $\forall z \in \mathbb{C}$
↑

$$(1-2i)(1+3i) = 7+i$$

Es: Sia $z = 2-3i$. Calcolare $\mathrm{Re}\left(\frac{1}{z}\right)$, $\mathrm{Im}\left(\frac{1}{z}\right)$

$$\frac{1}{z} = \frac{1}{2-3i} = \cancel{\frac{1}{2} - \frac{1}{3i}} \quad !!$$

$$\frac{1}{z} = \frac{1\,(2+3i)}{\underset{z}{(2-3i)}\,\underset{\bar z}{(2+3i)}} = \frac{2+3i}{13} = \frac{2}{13} + \frac{3}{13}i$$

$z\bar z = |z|^2$

$$\mathrm{Re}\left(\frac{1}{z}\right) = \frac{2}{13} \quad \mathrm{Im}\left(\frac{1}{z}\right) = \frac{3}{13}$$

$$z \neq 0 \quad \frac{1}{z} = \frac{\bar z}{z \cdot \bar z} = \frac{\bar z}{|z|^2} \leftarrow \odot$$

Es: Sia $z = 2-i$ Calcolare $\mathrm{Re}\left(\frac{1}{z^2}\right)$, $\mathrm{Im}\left(\frac{1}{z^2}\right)$

$$z^2 = (2-i)(2-i) = 4 + i^2 - 2i - 2i = 3-4i$$

$$\frac{1}{z^2} = \frac{1}{3-4i} = \frac{3+4i}{(3-4i)(3+4i)} = \frac{3}{25} + \frac{4}{25}i$$

Conclusione: $\mathrm{Re}\left(\frac{1}{z^2}\right) = \frac{3}{25}$ $\quad \mathrm{Im}\left(\frac{1}{z^2}\right) = \frac{4}{25}$

Eq. di 2° grado:

$\underbrace{a z^2 + bz + c = 0}$ $\qquad \underbrace{a, b, c \in \mathbb{R} \quad a \neq 0}$

$\Updownarrow$

$$\circledast \quad (z - z_0)^2 = \frac{\Delta}{4a^2} \qquad \left.\begin{array}{l} z_0 = -\dfrac{b}{2a} \\ \Delta = b^2 - 4ac \end{array}\right\} \swarrow \text{viste}$$

$\Delta = 0$ $\exists!$ soluzione $z = z_0$

$\Delta > 0 \Rightarrow$ 2 soluz. reali : $z - z_0 = \pm\frac{\sqrt{\Delta}}{2a}$ $\qquad z = \frac{-b \pm \sqrt{\Delta}}{2a}$

Se $\Delta < 0$, NON ci sono soluz. reali !!

PERÒ ABBIAMO 2 soluzioni in $\mathbb{C}$

$-\Delta > 0$

$$\circledast \iff (z - z_0)^2 = (-1)\underbrace{\frac{(-\Delta)}{4a^2}}_{>0}$$

ha soluzioni in $\mathbb{C}$: $z - z_0 = \pm i\frac{\sqrt{-\Delta}}{2a}$

cioè $z = \frac{-b \pm i\sqrt{-\Delta}}{2a}$ $\quad$ 2 soluz. in $\mathbb{C}$, una il coniugato dell'altra

Es: $\circledcirc\!\Longrightarrow$

Risolvere in $\mathbb{C}$: $\swarrow$ NON ha soluzioni in $\mathbb{R}$ $\qquad \Delta = b^2 - 4ac = -3 < 0$

$$\boxtimes \quad z^2 + z + 1 = 0$$

Abbiamo $z_1 = \frac{-1}{2} + \frac{i\sqrt{3}}{2}$ $\qquad z_2 = -\frac{1}{2} - \frac{i\sqrt{3}}{2}$ $\qquad z_2 = \overline{z_1}$

Verifichiamo che z_1 è soluzione di $\boxtimes$:

$$\underbrace{\left(\frac{-1}{2} + \frac{i\sqrt{3}}{2}\right)\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)}_{z_1^2} + \underbrace{-\frac{1}{2} + \frac{i\sqrt{3}}{2}}_{z_1} + 1 \overset{?}{=} 0$$

$$\frac{1}{4} - \frac{3}{4} - \frac{1}{4}i\sqrt{3} - \frac{1}{4}i\sqrt{3} - \frac{1}{2} + \frac{i\sqrt{3}}{2} + 1 =$$

$$-\frac{1}{2} \cancel{-\frac{1}{2}i\sqrt{3}} - \frac{1}{2} \cancel{+\frac{i\sqrt{3}}{2}} + 1 = 0 \qquad \underline{\text{O.K}}$$

Idem z_2 . . .

Risolvere in $\mathbb{C}$: $z^2 + z + 5 = 0$ $\qquad \Delta = -19$

verificare

$z_1 = -\frac{1}{2} + i\frac{\sqrt{19}}{2}$ $\qquad z_2 = \overline{z_1}$

Es: Rappresentare graficamente in $\mathbb{R}^2$ il seguente sottoinsieme di $\mathbb{C}$:

$$A = \{z \in \mathbb{C} : \underbrace{1 \le |z - i| \le 2}\}$$

$z = x + iy$ (generico n° complesso) $\updownarrow$

$z = a + ib$
$|z| = \sqrt{a^2+b^2}$

$1 \le |(x+iy) - i| \le 2$ cioè $1 \le |x + i(y-1)| \le 2$ cioè

$$1 \le \sqrt{x^2 + (y-1)^2} \le 2 \iff$$

↓ elevo al quadrato

$$\underbrace{1 \le x^2 + (y-1)^2 \le 4} \swarrow A$$

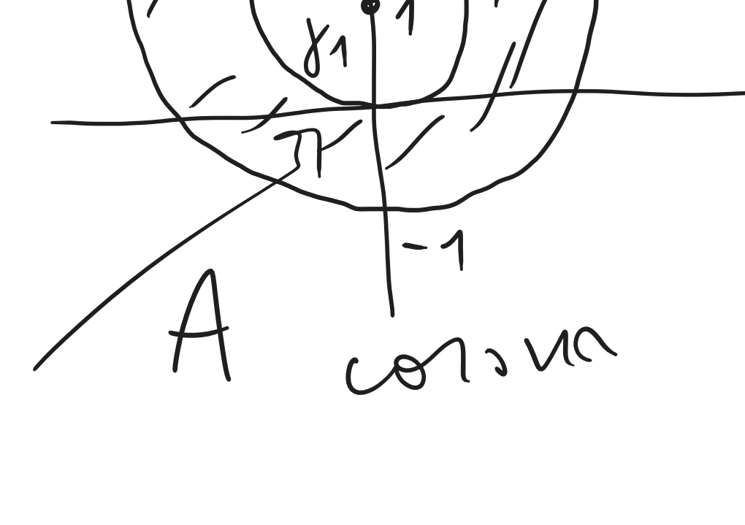
$$\begin{cases} 1 \le x^2 + (y-1)^2 \\ x^2 + (y-1)^2 \le 4 \end{cases}$$

$\gamma_1 : x^2 + (y-1)^2 = 1$

$\gamma_2 : x^2 + (y-1)^2 = 4$

punti esterni a γ_1

punti interni a γ_2



A corona circolare compresa tra γ_1 e γ_2